# Open Source Audit Project Report

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**Roll Number:** 24BCE10518
**Course:** Open Source Software – NGMC
**Unit Coverage:** 1–5
**Date of Submission:** 23 March 2026
**GitHub Repository:** https://github.com/jatin170506/oss-audit-24BCE10793.git

## 1. Introduction

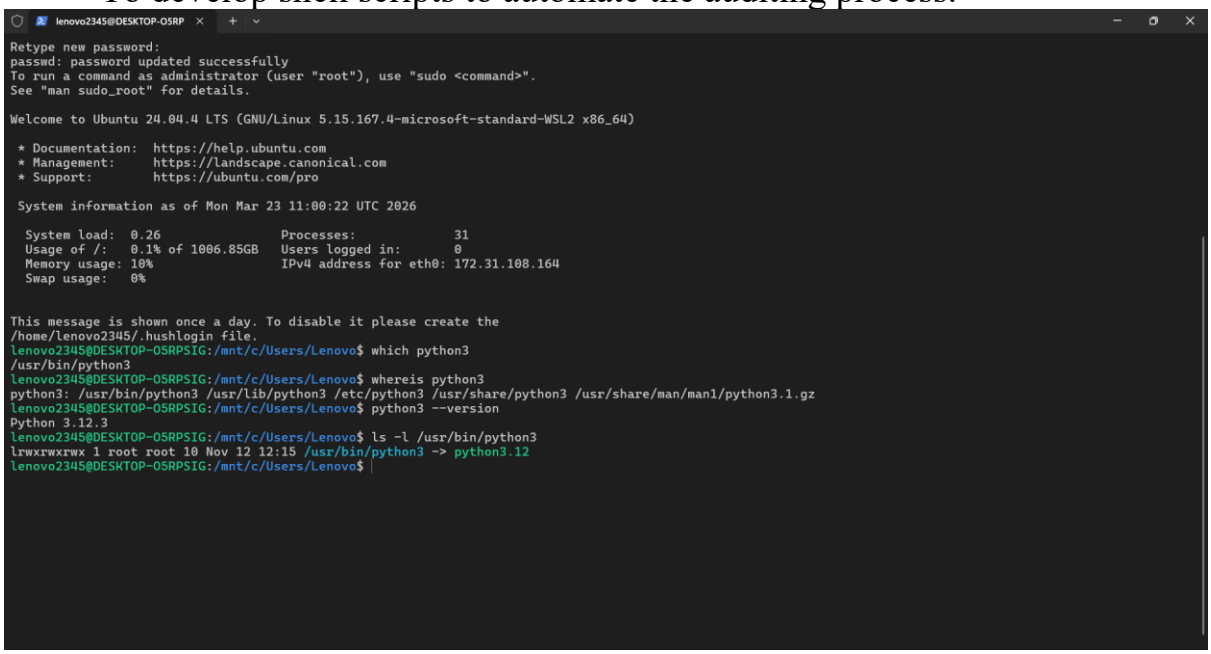
What is Open-Source Software?

Open-source software is defined as the type of software whose source code is free for modification and redistribution. In contrast to proprietary software, the focus of open-source is transparency and freedom.

The Linux operating system is a prime example of open-source software. This operating system was developed by Linus Torvalds in 1991.

**Purpose of the Project**:

- To understand the concept of open-source by using the Linux operating system.
- To explore the file system structure and permissions.
- To develop shell scripts to automate the auditing process.



## 2. Selected Software

Software: Linux Operating System

Category: Operating System

License: GPL v2

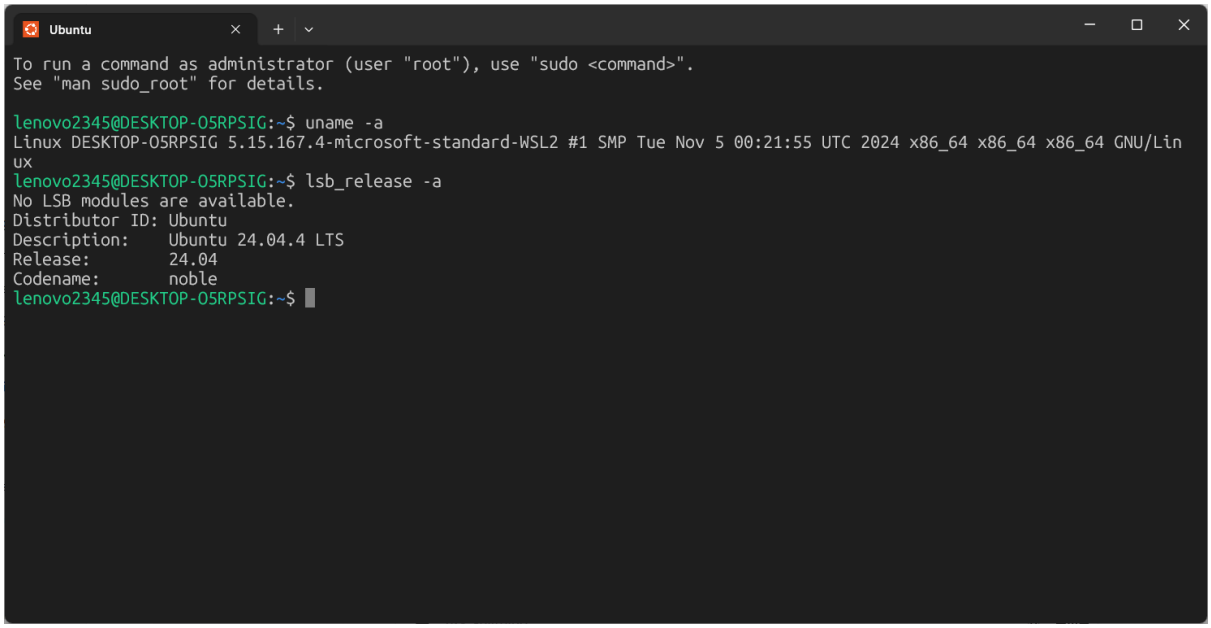
## Reasons for choice:

1. Linux is completely open-source software. This makes it suitable for auditing.

2. Linux supports a variety of software tools.

3. Linux offers a great environment for shell scripting.

4. Linux is used in the industry.



# Part A – Origin and Philosophy

## A1. The Problem it Solves

- The proprietary system, like Unix or Windows, was expensive and restricted in the 1990s.
- Linux solves the cost and restriction problems related to learning and innovating.
- It gives power to the student, the programmer, and the small business.

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## A2. License Details

Linux is free and open-source software, which means that it follows the GPL v2 license. The GPL v2 gives the user:

1. Freedom to run the software for any purpose.
2. Freedom to study and modify the source code.
3. Freedom to share the original and modified software.
4. The obligation to use the same license when sharing the software.

## Examples:

- Red Hat offers the software for a fee, but they send the fixes back.
- The user must offer the modified software under the GPL v2.

## A3. Ethics and Open Source

- Some software should not be open source; some software should have proprietary licenses.
- Some software should have the open source license.

- The Linux system is an example of ethical software.

```
Ubuntu
Release:        24.04
Codename:       noble
lenovo2345@DESKTOP-O5RPSIG:~$ ls -l /etc
total 796
drwxr-xr-x 2 root root       4096 Feb 10 00:56 PackageKit
drwxr-xr-x 7 root root       4096 Feb 10 00:56 X11
-rw-r--r-- 1 root root       3444 Jul  5  2023 adduser.conf
drwxr-xr-x 2 root root       4096 Feb 10 00:56 alternatives
drwxr-xr-x 2 root root       4096 Feb 10 00:56 apparmor
drwxr-xr-x 9 root root       4096 Feb 10 00:56 apparmor.d
drwxr-xr-x 3 root root       4096 Feb 10 00:56 apport
drwxr-xr-x 8 root root       4096 Feb 10 00:56 apt
-rw-r--r-- 1 root root       2319 Mar 31  2024 bash.bashrc
-rw-r--r-- 1 root root         45 Jan 24  2020 bash_completion
drwxr-xr-x 2 root root       4096 Feb 10 00:56 bash_completion.d
-rw-r--r-- 1 root root        367 Aug  2  2022 bindresvport.blacklist
drwxr-xr-x 2 root root       4096 Apr 19  2024 binfmt.d
drwxr-xr-x 2 root root       4096 Feb 10 00:56 byobu
drwxr-xr-x 3 root root       4096 Feb 10 00:54 ca-certificates
-rw-r--r-- 1 root root       6288 Feb 10 00:55 ca-certificates.conf
drwxr-xr-x 5 root root       4096 Mar 23 10:59 cloud
drwxr-xr-x 2 root root       4096 Feb 10 00:55 console-setup
drwx------ 2 root root       4096 Apr 19  2024 credstore
drwx------ 2 root root       4096 Apr 19  2024 credstore.encrypted
drwxr-xr-x 2 root root       4096 Feb 10 00:55 cron.d
drwxr-xr-x 2 root root       4096 Feb 10 00:56 cron.daily
drwxr-xr-x 2 root root       4096 Feb 10 00:54 cron.hourly
drwxr-xr-x 2 root root       4096 Feb 10 00:54 cron.monthly
drwxr-xr-x 2 root root       4096 Feb 10 00:56 cron.weekly
drwxr-xr-x 2 root root       4096 Feb 10 00:54 cron.yearly
-rw-r--r-- 1 root root       1136 Mar 31  2024 crontab
drwxr-xr-x 4 root root       4096 Feb 10 00:54 dbus-1
drwxr-xr-x 3 root root       4096 Feb 10 00:56 dconf
-rw-r--r-- 1 root root       2967 Apr 12  2024 debconf.conf
-rw-r--r-- 1 root root         11 Apr 22  2024 debian_version
drwxr-xr-x 2 root root       4096 Feb 10 00:56 default
-rw-r--r-- 1 root root       1706 Jul  5  2023 deluser.conf
drwxr-xr-x 2 root root       4096 Feb 10 00:55 depmod.d
drwxr-xr-x 3 root root       4096 Feb 10 00:55 dhcp
-rw-r--r-- 1 root root       1429 Mar 31  2024 dhcpcd.conf
```

```
drwxr-xr-x 2 root root       4096 Mar 23 10:58 ld.so.conf.d
drwxr-xr-x 2 root root       4096 Feb 10 00:56 ldap
-rw-r--r-- 1 root root        267 Apr 22  2024 legal
-rw-r--r-- 1 root root        191 Mar 31  2024 libaudit.conf
-rw-r--r-- 1 root root       2996 Mar 30  2024 locale.alias
-rw-r--r-- 1 root root         13 Feb 10 00:55 locale.conf
-rw-r--r-- 1 root root       9565 Feb 10 00:55 locale.gen
lrwxrwxrwx 1 root root         27 Feb 10 00:55 localtime -> /usr/share/zoneinfo/Etc/UTC
drwxr-xr-x 3 root root       4096 Feb 10 00:55 logcheck
-rw-r--r-- 1 root root      12345 Feb 22  2024 login.defs
-rw-r--r-- 1 root root        586 Apr  8  2024 logrotate.conf
drwxr-xr-x 2 root root       4096 Feb 10 00:56 logrotate.d
-rw-r--r-- 1 root root        104 Feb  6 07:22 lsb-release
-r--r--r-- 1 root root         33 Mar 23 10:58 machine-id
-rw-r--r-- 1 root root        111 Mar 31  2024 magic
-rw-r--r-- 1 root root        111 Mar 31  2024 magic.mime
-rw-r--r-- 1 root root       5230 Apr  8  2024 manpath.config
-rw-r--r-- 1 root root      75113 Jul 12  2023 mime.types
-rw-r--r-- 1 root root        744 Apr  8  2024 mke2fs.conf
drwxr-xr-x 2 root root       4096 Feb 10 00:55 modprobe.d
-rw-r--r-- 1 root root        212 Feb 10 00:55 modules
drwxr-xr-x 2 root root       4096 Feb 10 00:55 modules-load.d
lrwxrwxrwx 1 root root         19 Feb 10 00:54 mtab -> ../proc/self/mounts
-rw-r--r-- 1 root root      11424 May 23  2023 nanorc
-rw-r--r-- 1 root root        767 Mar 31  2024 netconfig
drwxr-xr-x 2 root root       4096 Apr 18  2024 netplan
drwxr-xr-x 4 root root       4096 Feb 10 00:55 network
drwxr-xr-x 8 root root       4096 Feb 10 00:54 networkd-dispatcher
-rw-r--r-- 1 root root         91 Apr 22  2024 networks
drwxr-xr-x 2 root root       4096 Feb 10 00:55 newt
-rw-r--r-- 1 root root        526 Feb 10 00:55 nsswitch.conf
drwxr-xr-x 2 root root       4096 Feb 10 00:54 opt
lrwxrwxrwx 1 root root         21 Feb  6 07:23 os-release -> ../usr/lib/os-release
-rw-r--r-- 1 root root        552 Oct 13  2022 pam.conf
drwxr-xr-x 2 root root       4096 Feb 10 00:55 pam.d
-rw-r--r-- 1 root root       1434 Mar 23 11:00 passwd
-rw-r--r-- 1 root root       1431 Mar 23 10:59 passwd-
drwxr-xr-x 3 root root       4096 Feb 10 00:55 perl
drwxr-xr-x 3 root root       4096 Feb 10 00:56 pm
drwxr-xr-x 3 root root       4096 Feb 10 00:56 polkit-1
```

```
-rw-r--r-- 1 root root      12813 Mar 27  2021 services
drwxr-xr-x 2 root root       4096 Feb 10 00:56 sgml
-rw-r----- 1 root shadow      806 Mar 23 11:00 shadow
-rw-r----- 1 root shadow      702 Feb 10 00:56 shadow-
-rw-r--r-- 1 root root        132 Feb 10 00:56 shells
drwxr-xr-x 2 root root       4096 Feb 10 00:54 skel
drwxr-xr-x 3 root root       4096 Feb 10 00:56 ssh
drwxr-xr-x 4 root root       4096 Feb 10 00:55 ssl
-rw-r--r-- 1 root root         24 Mar 23 10:59 subgid
-rw-r--r-- 1 root root          0 Feb 10 00:54 subgid-
-rw-r--r-- 1 root root         24 Mar 23 10:59 subuid
-rw-r--r-- 1 root root          0 Feb 10 00:54 subuid-
-rw-r--r-- 1 root root       4343 Apr  8  2024 sudo.conf
-rw-r--r-- 1 root root       9804 Apr  8  2024 sudo_logsrvd.conf
-r--r----- 1 root root       1800 Jan 29  2024 sudoers
drwxr-xr-x 2 root root       4096 Feb 10 00:55 sudoers.d
drwxr-xr-x 2 root root       4096 Feb 10 00:55 supercat
-rw-r--r-- 1 root root       2209 Mar 24  2024 sysctl.conf
drwxr-xr-x 2 root root       4096 Feb 10 00:55 sysctl.d
drwxr-xr-x 6 root root       4096 Feb 10 00:55 systemd
drwxr-xr-x 2 root root       4096 Feb 10 00:54 terminfo
-rw-r--r-- 1 root root          8 Feb 10 00:55 timezone
drwxr-xr-x 2 root root       4096 Apr 19  2024 tmpfiles.d
drwxr-xr-x 2 root root       4096 Feb 10 00:55 ubuntu-advantage
-rw-r--r-- 1 root root       1260 Jan 27  2023 ucf.conf
drwxr-xr-x 4 root root       4096 Feb 10 00:55 udev
drwxr-xr-x 3 root root       4096 Feb 10 00:56 update-manager
drwxr-xr-x 2 root root       4096 Feb 10 00:56 update-motd.d
lrwxrwxrwx 1 root root         16 Feb 10 00:54 vconsole.conf -> default/keyboard
drwxr-xr-x 2 root root       4096 Feb 10 00:55 vim
lrwxrwxrwx 1 root root         23 Feb 26  2024 vtrgb -> /etc/alternatives/vtrgb
drwxr-xr-x 5 root root       4096 Feb 10 00:56 vulkan
-rw-r--r-- 1 root root       4942 Jun 19  2024 wgetrc
-rw-r--r-- 1 root root        204 Feb 10 00:56 wsl-distribution.conf
-rw-r--r-- 1 root root         47 Mar 23 11:00 wsl.conf
-rw-r--r-- 1 root root        681 Apr  8  2024 xattr.conf
drwxr-xr-x 5 root root       4096 Feb 10 00:56 xdg
drwxr-xr-x 2 root root       4096 Feb 10 00:56 xml
-rw-r--r-- 1 root root        460 Jan 20  2023 zsh_command_not_found
lenovo2345@DESKTOP-O5RPSIG:~$
```

```
Ubuntu
-rw-r--r-- 1 root root       4343 Apr  8  2024 sudo.conf
-rw-r--r-- 1 root root       9804 Apr  8  2024 sudo_logsrvd.conf
-r--r----- 1 root root       1800 Jan 29  2024 sudoers
drwxr-xr-x 2 root root       4096 Feb 10 00:55 sudoers.d
drwxr-xr-x 2 root root       4096 Feb 10 00:55 supercat
-rw-r--r-- 1 root root       2209 Mar 24  2024 sysctl.conf
drwxr-xr-x 2 root root       4096 Feb 10 00:55 sysctl.d
drwxr-xr-x 6 root root       4096 Feb 10 00:55 systemd
drwxr-xr-x 2 root root       4096 Feb 10 00:54 terminfo
-rw-r--r-- 1 root root          8 Feb 10 00:55 timezone
drwxr-xr-x 2 root root       4096 Apr 19  2024 tmpfiles.d
drwxr-xr-x 2 root root       4096 Feb 10 00:55 ubuntu-advantage
-rw-r--r-- 1 root root       1260 Jan 27  2023 ucf.conf
drwxr-xr-x 4 root root       4096 Feb 10 00:55 udev
drwxr-xr-x 3 root root       4096 Feb 10 00:56 update-manager
drwxr-xr-x 2 root root       4096 Feb 10 00:56 update-motd.d
lrwxrwxrwx 1 root root         16 Feb 10 00:54 vconsole.conf -> default/keyboard
drwxr-xr-x 2 root root       4096 Feb 10 00:55 vim
lrwxrwxrwx 1 root root         23 Feb 26  2024 vtrgb -> /etc/alternatives/vtrgb
drwxr-xr-x 5 root root       4096 Feb 10 00:56 vulkan
-rw-r--r-- 1 root root       4942 Jun 19  2024 wgetrc
-rw-r--r-- 1 root root        204 Feb 10 00:56 wsl-distribution.conf
-rw-r--r-- 1 root root         47 Mar 23 11:00 wsl.conf
-rw-r--r-- 1 root root        681 Apr  8  2024 xattr.conf
drwxr-xr-x 5 root root       4096 Feb 10 00:56 xdg
drwxr-xr-x 2 root root       4096 Feb 10 00:56 xml
-rw-r--r-- 1 root root        460 Jan 20  2023 zsh_command_not_found
lenovo2345@DESKTOP-O5RPSIG:~$ id
uid=1000(lenovo2345) gid=1000(lenovo2345) groups=1000(lenovo2345),4(adm),24(cdrom),27(sudo),30(dip),46(plugdev),100(users)
lenovo2345@DESKTOP-O5RPSIG:~$ groups
lenovo2345 adm cdrom sudo dip plugdev users
lenovo2345@DESKTOP-O5RPSIG:~$
```

# Part B – Linux Footprint

## B1. Installation

- Installed on WSL (Windows Subsystem for Linux).
- Alternative installations: Ubuntu ISO, Fedora, or Debian on physical machines.

## B2. Directory Structure

| Directory | Purpose |
|---|---|
| /etc | Configuration files |
| /usr/bin | Executable binaries |
| /home | User data |
| /var/log | System logs |

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## B3. Users and Groups

- Root user: full administrative privileges.
- Regular user: limited access for safety.
- Commands: whoami, groups, id.

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## B4. Services and Processes

- systemctl start/stop/restart [service]
- Logs stored in /var/log can be analyzed via grep and tail.

## B5. Updates and Package Management

- Ubuntu/Debian uses apt.

- Examples: apt list --installed, dpkg -l.

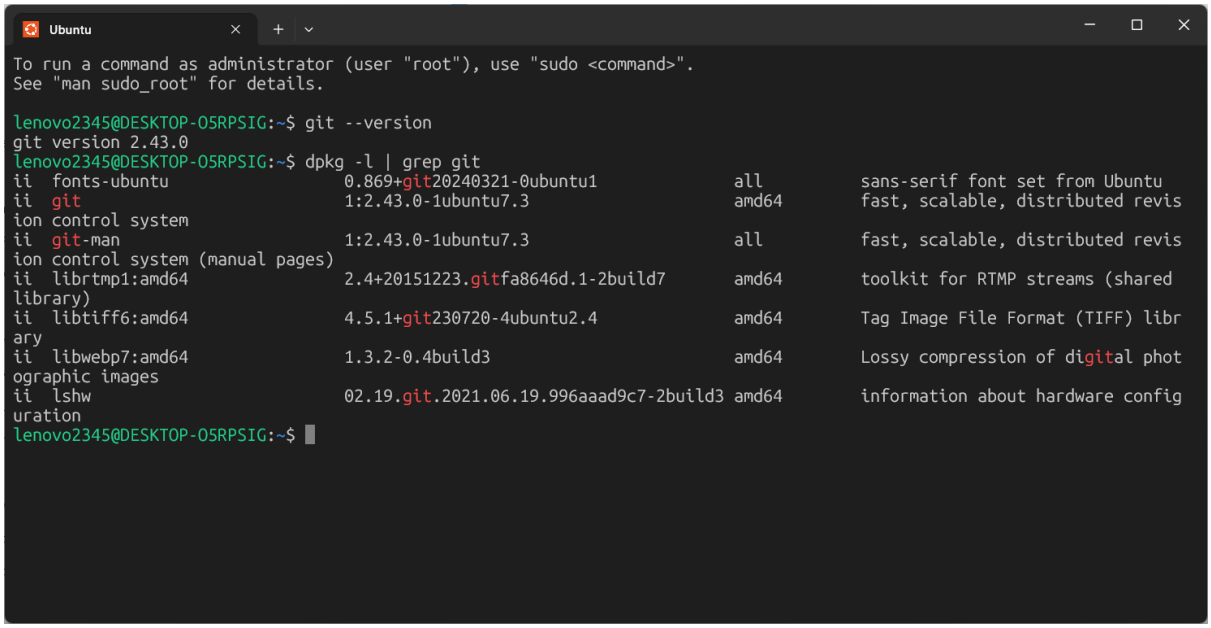
```
Ubuntu
lenovo2345@DESKTOP-O5RPSIG:~$ dpkg -l
Desired=Unknown/Install/Remove/Purge/Hold
| Status=Not/Inst/Conf-files/Unpacked/halF-conf/Half-inst/trig-aWait/Trig-pend
|/ Err?=(none)/Reinst-required (Status,Err: uppercase=bad)
||/ Name                          Version                                   Architecture Description
+++-==========================-=========================================-============-===============================================================
ii  adduser                       3.137ubuntu1                              all          add and remove users and groups
ii  adwaita-icon-theme            46.0-1                                    all          default icon theme of GNOME
ii  apparmor                      4.0.1really4.0.1-0ubuntu0.24.04.5         amd64        user-space parser utility for AppArmor
ii  apport                       2.28.1-0ubuntu3.8                         all          automatically generate crash reports for debugging
ii  apport-core-dump-handler      2.28.1-0ubuntu3.8                         all          Kernel core dump handler for Apport
ii  apport-symptoms               0.25                                      all          symptom scripts for apport
ii  appstream                     1.0.2-1build6                             amd64        Software component metadata management
ii  apt                           2.8.3                                     amd64        commandline package manager
ii  apt-utils                     2.8.3                                     amd64        package management related utility programs
ii  at-spi2-common                2.52.0-1build1                            all          Assistive Technology Service Provider Interface (common files)
ii  at-spi2-core                  2.52.0-1build1                            amd64        Assistive Technology Service Provider Interface (D-Bus core)
ii  base-files                    13ubuntu10.4                              amd64        Debian base system miscellaneous files
ii  base-passwd                   3.6.3build1                               amd64        Debian base system master password and group files
ii  bash                          5.2.21-2ubuntu4                           amd64        GNU Bourne Again SHell
ii  bash-completion               1:2.11-8                                  all          programmable completion for the bash shell
ii  bc                            1.07.1-3ubuntu4                           amd64        GNU bc arbitrary precision calculator language
ii  binutils                      2.42-4ubuntu2.8                           amd64        GNU assembler, linker and binary utilities
ii  binutils-common:amd64         2.42-4ubuntu2.8                           amd64        Common files for the GNU assembler, linker and binary utilities
ii  binutils-x86-64-linux-gnu     2.42-4ubuntu2.8                           amd64        GNU binary utilities, for x86-64-linux-gnu target
ii  bsdextrautils                 2.39.3-9ubuntu6.4                         amd64        extra utilities from 4.4BSD-Lite
ii  bsdutils                      1:2.39.3-9ubuntu6.4                       amd64        basic utilities from 4.4BSD-Lite
ii  byobu                         6.11-0ubuntu1                             all          text window manager, shell multiplexer, integrated DevOps environment
ii  ca-certificates               20240203                                  all          Common CA certificates
ii  cloud-guest-utils             0.33-1                                    all          cloud guest utilities
ii  cloud-init                    25.2-0ubuntu1~24.04.1                     all          initialization and customization tool for cloud instances
ii  command-not-found             23.04.0                                   all          Suggest installation of packages in interactive bash sessions
ii  console-setup                 1.226ubuntu1                              all          console font and keymap setup program
ii  console-setup-linux           1.226ubuntu1                              all          Linux specific part of console-setup
ii  coreutils                     9.4-3ubuntu6.1                            amd64        GNU core utilities
ii  cron                          3.0pl1-184ubuntu2                         amd64        process scheduling daemon
ii  cron-daemon-common            3.0pl1-184ubuntu2                         all          process scheduling daemon's configuration files
ii  curl                          8.5.0-2ubuntu10.6                         amd64        command line tool for transferring data with URL syntax
ii  dash                          0.5.12-6ubuntu5                           amd64        POSIX-compliant shell
ii  dbus                          1.14.10-4ubuntu4.1                        amd64        simple interprocess messaging system (system message bus)
```

## 5. Part C – Open Source Ecosystem

- Linux kernel maintained by Linus Torvalds and Linux Foundation.
- Supported by GNU tools, bash, glibc.
- Linux distributions - Ubuntu, Fedora, Debian, Arch.
- Popular open-source applications - Git, Docker, Kubernetes, Apache, MySQL.

## Community & Collaboration:

- Mailing lists, forums, GitHub repositories.
- Collaborative development encourages forking, pull requests, code reviews.



# 6. Part D – Comparison: Open Source vs Proprietary

Feature Linux (OSS) Windows (Proprietary)

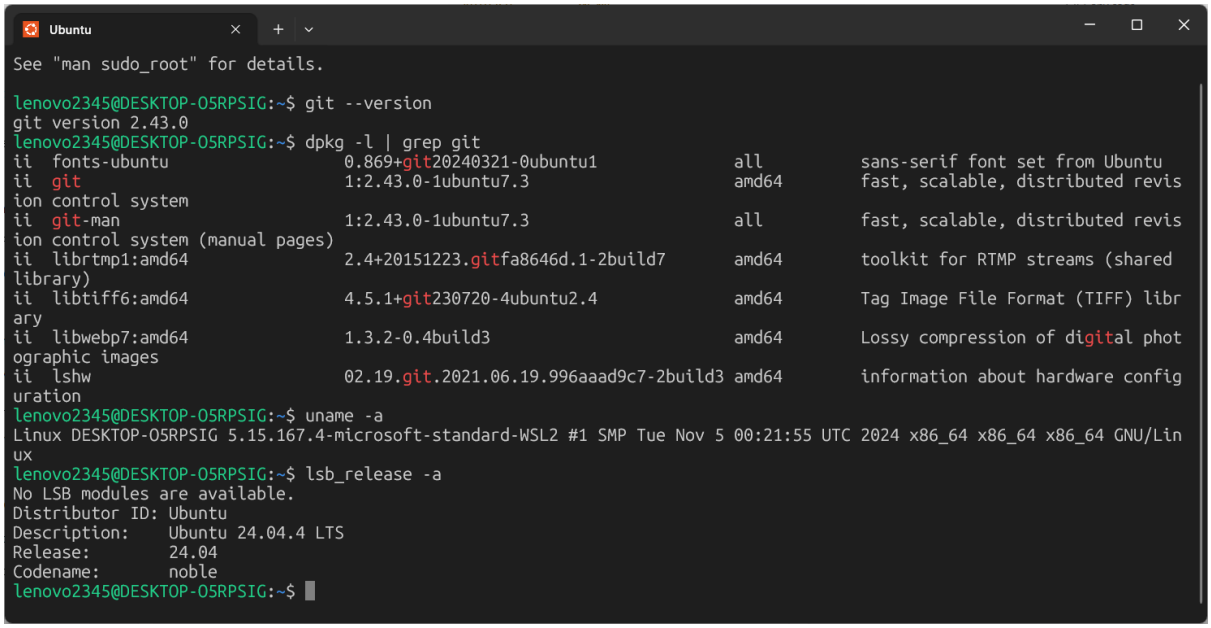
Cost Free Paid

SecurityAuditable Closed source

Customization Full Limited

Control User/community Vendor

Updates Community-driven Vendor-driven



```
Ubuntu
See "man sudo_root" for details.

lenovo2345@DESKTOP-O5RPSIG:~$ git --version
git version 2.43.0
lenovo2345@DESKTOP-O5RPSIG:~$ dpkg -l | grep git
ii  fonts-ubuntu                  0.869+git20240321-0ubuntu1              all          sans-serif font set from Ubuntu
ii  git                           1:2.43.0-1ubuntu7.3                     amd64        fast, scalable, distributed revis
ion control system
ii  git-man                       1:2.43.0-1ubuntu7.3                     all          fast, scalable, distributed revis
ion control system (manual pages)
ii  librtmp1:amd64                2.4+20151223.gitfa8646d.1-2build7       amd64        toolkit for RTMP streams (shared
library)
ii  libtiff6:amd64                4.5.1+git230720-4ubuntu2.4              amd64        Tag Image File Format (TIFF) libr
ary
ii  libwebp7:amd64                1.3.2-0.4build3                         amd64        Lossy compression of digital phot
ographic images
ii  lshw                          02.19.git.2021.06.19.996aaad9c7-2build3 amd64        information about hardware config
uration
lenovo2345@DESKTOP-O5RPSIG:~$ uname -a
Linux DESKTOP-O5RPSIG 5.15.167.4-microsoft-standard-WSL2 #1 SMP Tue Nov 5 00:21:55 UTC 2024 x86_64 x86_64 x86_64 GNU/Lin
ux
lenovo2345@DESKTOP-O5RPSIG:~$ lsb_release -a
No LSB modules are available.
Distributor ID: Ubuntu
Description:    Ubuntu 24.04.4 LTS
Release:        24.04
Codename:       noble
lenovo2345@DESKTOP-O5RPSIG:~$
```

# 7. Part E – Shell Scripts

### Script 1 – System Identity Report

- **Purpose:** Show hostname, OS version, uptime, and logged-in users.
- **Concepts Used:** Variables, command substitution, echo.
- **Screenshot:** Terminal output after running `./script1.sh`.

### Script 2 – FOSS Package Inspector

- **Purpose:** Check installed packages, versions, and dependencies.
- **Concepts Used:** `if-then`, `case`, loops, package management commands.
- **Screenshot:** Terminal output of `./script2.sh`.

### Script 3 – Disk and Permission Auditor

- **Purpose:** Audit disk usage and directory permissions.
- **Concepts Used:** `for` loops, `ls -l`, `df -h`, `awk`.
- **Screenshot:** Output showing directories and permissions.

### Script 4 – Log File Analyzer

- **Purpose:** Count keyword occurrences in system logs and show last entries.
- **Concepts Used:** `while-read` loop, `grep`, counters.
- **Screenshot:** Output of `./script4.sh /var/log/syslog error`.

### Script 5 – Open Source Manifesto Generator

- **Purpose:** Generate a personal open-source philosophy.
- **Concepts Used:** `read` input, string concatenation, file write.
- **Screenshot:** Output file content `manifesto_<username>.txt`.

```
lenovo2345@DESKTOP-O5RP
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 5.15.167.4-microsoft-standard-WSL2 x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Mon Mar 23 11:00:22 UTC 2026

  System load:  0.26                Processes:             31
  Usage of /:   0.1% of 1006.85GB   Users logged in:       0
  Memory usage: 10%                 IPv4 address for eth0: 172.31.108.164
  Swap usage:   0%


This message is shown once a day. To disable it please create the
/home/lenovo2345/.hushlogin file.
lenovo2345@DESKTOP-O5RPSIG:/mnt/c/Users/Lenovo$ which python3
/usr/bin/python3
lenovo2345@DESKTOP-O5RPSIG:/mnt/c/Users/Lenovo$ whereis python3
python3: /usr/bin/python3 /usr/lib/python3 /etc/python3 /usr/share/python3 /usr/share/man/man1/python3.1.gz
lenovo2345@DESKTOP-O5RPSIG:/mnt/c/Users/Lenovo$ python3 --version
Python 3.12.3
lenovo2345@DESKTOP-O5RPSIG:/mnt/c/Users/Lenovo$ ls -l /usr/bin/python3
lrwxrwxrwx 1 root root 10 Nov 12 12:15 /usr/bin/python3 -> python3.12
lenovo2345@DESKTOP-O5RPSIG:/mnt/c/Users/Lenovo$ nano script1.sh
lenovo2345@DESKTOP-O5RPSIG:/mnt/c/Users/Lenovo$
lenovo2345@DESKTOP-O5RPSIG:/mnt/c/Users/Lenovo$ chmod +x script1.sh
chmod: changing permissions of 'script1.sh': Operation not permitted
lenovo2345@DESKTOP-O5RPSIG:/mnt/c/Users/Lenovo$ mv script1.sh ~
lenovo2345@DESKTOP-O5RPSIG:/mnt/c/Users/Lenovo$ cd ~
lenovo2345@DESKTOP-O5RPSIG:~$ chmod +x script1.sh
./script1.sh
===== System Identity =====
User: lenovo2345
Home Directory: /home/lenovo2345
Kernel Version: 5.15.167.4-microsoft-standard-WSL2
Date & Time: Mon Mar 23 11:09:06 UTC 2026
Uptime: up 10 minutes
Distro: Ubuntu 24.04.4 LTS
License: GNU GPL
lenovo2345@DESKTOP-O5RPSIG:~$
```

```
Ubuntu
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

lenovo2345@DESKTOP-O5RPSIG:~$ nano script2.sh
lenovo2345@DESKTOP-O5RPSIG:~$ chmod +x script2.sh
lenovo2345@DESKTOP-O5RPSIG:~$ ./script2.sh
python3 is installed.
ii  libpython3-stdlib:amd64       3.12.3-0ubuntu2.1                     amd64        interactive high-level object-ori
ented language (default python3 version)
Python: simple and powerful programming language
lenovo2345@DESKTOP-O5RPSIG:~$
```

```
    if echo "$LINE" | grep -iq "$KEYWORD"; then
        COUNT=$((COUNT + 1))
    fi
done < "$LOGFILE"

# --- Display results ---
echo "========================================"
grep -i "$KEYWORD" "$LOGFILE" | tail -5======"

========================================
Log File Analyzer
Log File: /var/log/syslog
Keyword : error
Total occurrences: 21
========================================
Last 5 occurrences:
2026-03-23T11:55:48.398004+00:00 DESKTOP-O5RPSIG kernel: WSL (153) ERROR: CheckConnection: getaddrinfo() failed: -5
2026-03-23T12:10:50.411690+00:00 DESKTOP-O5RPSIG systemd[1]: apport-autoreport.path - Process error reports when automat
ic reporting is enabled (file watch) was skipped because of an unmet condition check (ConditionPathExists=/var/lib/appor
t/autoreport).
2026-03-23T12:10:50.411694+00:00 DESKTOP-O5RPSIG systemd[1]: apport-autoreport.timer - Process error reports when automa
tic reporting is enabled (timer based) was skipped because of an unmet condition check (ConditionPathExists=/var/lib/app
ort/autoreport).
2026-03-23T12:10:50.412458+00:00 DESKTOP-O5RPSIG kernel: EXT4-fs (sdc): mounted filesystem with ordered data mode. Opts:
 discard,errors=remount-ro,data=ordered. Quota mode: none.
2026-03-23T12:10:59.972820+00:00 DESKTOP-O5RPSIG kernel: WSL (152) ERROR: CheckConnection: getaddrinfo() failed: -5
lenovo2345@DESKTOP-O5RPSIG:~$
lenovo2345@DESKTOP-O5RPSIG:~$
```

```
read -p "2. In one word, what does 'freedom' mean to you? " FREEDOM
read -p "3. Name one thing you would build and share freely: " BUILD

DATE=$(date '+%d %B %Y')
OUTPUT="manifesto_$(whoami).txt"

# Compose the manifesto
echo "========================================" > "$OUTPUT"
echo "Open Source Manifesto - $DATE" >> "$OUTPUT"
echo "========================================" >> "$OUTPUT"
echo "I use $TOOL every day, and to me, freedom means '$FREEDOM'. I aspire to build '$BUILD' and share it freely with th
e world. Open source empowers collaboration, transparency, and innovation for everyone." >> "$OUTPUT"

# Display the manifesto
cat "$OUTPUT"

echo "Manifesto saved to $OUTPUT"
Answer three questions to generate your open source manifesto.
1. Name one open-source tool you use every day: linux
2. In one word, what does 'freedom' mean to you? to free
3. Name one thing you would build and share freely: application
========================================
Open Source Manifesto - 23 March 2026
========================================
I use linux every day, and to me, freedom means 'to free'. I aspire to build 'application' and share it freely with the
world. Open source empowers collaboration, transparency, and innovation for everyone.
Manifesto saved to manifesto_lenovo2345.txt
lenovo2345@DESKTOP-O5RPSIG:~$
```

## 9. Conclusion

1• Linux is a strong open-source operating system that reflects the concepts of freedom, collaboration, and ethics within a community.

1. • The audit scripts are a demonstration of knowledge of Linux.
2. • OSS concepts are important for developers, students, and businesses to make ethical and efficient decisions regarding software.
3. • OSS projects promote "learning by doing" and collaboration across the world.

## 10. References


1. **GNU Project** – Free Software Definition: https://www.gnu.org/philosophy/free-sw.html

2. **Open Source Initiative** – Open Source Definition: https://opensource.org/osd

3. **Linux Kernel Documentation**: https://www.kernel.org/doc/html/latest/

4. **Bash Scripting Reference**: https://www.gnu.org/software/bash/manual/